5.2.17

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Question

Solve the given system of linear equations:

$$x - 3y - 7 = 0$$
$$3x - 3y - 15 = 0$$

Solution

The given equations of line can be expressed as:

$$\mathbf{n}^{\mathbf{T}}\mathbf{x} = c \tag{1}$$

$$\begin{pmatrix} 1 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 7 \tag{2}$$

$$\begin{pmatrix} 3 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 15 \tag{3}$$

The matrix we get by the above equations is as follows:

$$\begin{pmatrix} 1 & -3 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 15 \end{pmatrix} \tag{4}$$

Solution

The following is the augmented matrix which can be solved using Gaussian elimination.

$$\left(\begin{array}{cc|c} 1 & -3 & 7 \\ 3 & -3 & 15 \end{array}\right) \xrightarrow{R_2 \longrightarrow R_2 - 3R_1} \left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 6 & -6 \end{array}\right) \tag{5}$$

$$\left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 6 & -6 \end{array}\right) \xrightarrow{R_2 \longrightarrow \frac{R_2}{6}} \left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 1 & -1 \end{array}\right) \tag{6}$$

$$\left(\begin{array}{cc|c} 1 & -3 & 7 \\ 0 & 1 & -1 \end{array}\right) \xrightarrow{R_1 \longrightarrow R_1 + 3R_2} \left(\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & -1 \end{array}\right) \tag{7}$$

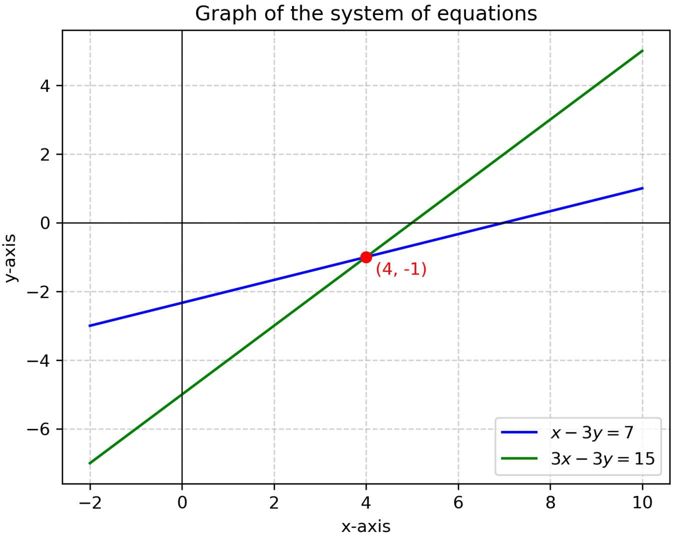
Solution

From the matrix,

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \tag{8}$$

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \tag{9}$$

Plot


Graph of the system of equations
y-axis
x-axis
(4, -1)
$x - 3y = 7$
$3x - 3y = 15$
4
2
0
−2
−4
−6
−2
0
2
4
6
8
10

C Code

```c
    #include <stdio.h>

// Solve system of equations:
// x - 3y = 7
// 3x - 3y = 15
//
// Returns solution in sol[0] = x, sol[1] = y
void solve_system(double sol[2]) {
    double a11 = 1, a12 = -3, b1 = 7;
    double a21 = 3, a22 = -3, b2 = 15;

    double det = a11 * a22 - a12 * a21;

    sol[0] = (b1 * a22 - a12 * b2) / det; // x
    sol[1] = (a11 * b2 - b1 * a21) / det; // y
}
```

Python Code

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Load the shared library
lib = ctypes.CDLL(./code.so)

# Define argument and return types
lib.solve_system.argtypes = [np.ctypeslib.ndpointer(dtype=np.
    double, ndim=1, flags=C_CONTIGUOUS)]
lib.solve_system.restype = None

 # Prepare array to hold solution [x, y]
sol = np.zeros(2, dtype=np.double)
```

Python Code

```
# Call the C function
lib.solve_system(sol)

# Print result
print(Solution of system:)
print(x =, sol[0])
print(y =, sol[1])

# Define the two equations in terms of y
# x - 3y = 7 => y = (x - 7)/3
# 3x - 3y = 15 => y = x - 5
def line1(x):
    return (x - 7) / 3

def line2(x):
    return x - 5
```

Python Code

```
# Solve intersection point (already known: x=4, y=-1)
intersection = (4, -1)

# Generate x values
x_vals = np.linspace(-2, 10, 100)

# Plot the lines
plt.plot(x_vals, line1(x_vals), label=r$x - 3y = 7$, color=blue)
plt.plot(x_vals, line2(x_vals), label=r$3x - 3y = 15$, color=green)

# Plot intersection point
plt.scatter(*intersection, color=red, zorder=5)
plt.text(intersection[0]+0.2, intersection[1]-0.5, f({intersection[0]}, {intersection[1]}), color=red)
```

Python Code

```
# Axis labels
plt.xlabel(x-axis)
plt.ylabel(y-axis)
plt.title(Graph of the system of equations)

# Grid, legend
plt.axhline(0, color=black, linewidth=0.7)
plt.axvline(0, color=black, linewidth=0.7)
plt.grid(True, linestyle=--, alpha=0.6)
plt.legend()

# Save the figure
plt.savefig(LE.png, dpi=300, bbox_inches=tight)

# Show plot
plt.show()
```